

Jacob Lynn

 Jacob Lynn |  jacob.lynn280@gmail.com |  +949-691-4585

EDUCATION

Stanford University

Class of 2029

Mechanical Engineering Major/Computer Science Minor

Relevant Coursework:

CS229: Intro to Robotics, ME210: Intro to Mechatronics (both graduate level courses)

RELEVANT EXPERIENCE

ARMLab Researcher (Assistive Robotics and Manipulation)

April 2026 - Present

- Worked 40hr weeks for 10 weeks over the summer and continuing in the lab during the school year
- Designing a electromagnetic linear actuator as a robotic muscle for tendon-driven robot hands with full position control (using PID) and force control (using PWM)
- Created a two-phase control policy using h-bridges
- Coded multiple ROS2 environments and a force simulation to determine electromagnet spacing
- Designed a original sensor that calculates the position the actuator by measuring inductance
- ARMLab applied for a patent of our current prototype

Stanford Student Space Institute - Europa Team Colead

Sep 2025 - Present

- Leading a team of ~10 people to design a low-cost, open source benthic lander that collects sediment, water, DNA, and video from ocean trench depths (9000m+)
- Leading the mechanical design which includes a custom soil scoop, nitskin tube, and pressure-based DNA collector
- Designing spring and galvanic timer based mechanisms because the extreme pressure does not allow electronics
- Completed multiple successful tests in the SF Bay and in Monterey in collaboration with Monterey Bay Aquarium Research Institute

Stanford Student Space Institute - IREC Team, Payload Subteam

Sep 2025 - Present

- Competing in the International Rocket Engineering Competition (IREC)
- Led the redesign of a L3 rocket payload that synthesizes crystals in microgravity for pharmaceutical testing
- Used a combination of laser cut metal and 3D printed interiors for a compact yet strong payload that fit within a 1U cubesat size

PROJECTS

Generic Limbed Robot/Robot Dog

[Link](#)

- Designed a custom QDD Actuator with a 5:1 ratio to power a full-scale 12DoF robot dog
- Designed multiple iterations of the robot dog over the period of a year
- Built

SKILLS

General Engineering	CAD, CAM, and FEA analysis in Fusion 360
Fabrication	CNC Machining, 3D Printing, General Workshop/Woodworking, Laser Cutting, Carbon Fiber/Composite Epoxy Layups
Computer Science	Python, JavaScript, Java, C, C++, ROS2